

GRAINGUARD · DESIGN DECISIONS REGISTER

GrainGuard design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-02	Amish Chadha	Follow-ups carried out (20 W panel and adjustable-disconnect controller in the BOM and model; cost USD 406.50); one new open decision, the stay-off service limit
0.2	2026-10-02	Amish Chadha	Amish approved the recommendations of open items 1 to 8 on 2026-10-02; all moved to decisions made; panel and charge controller lines to confirm updated
0.1	2026-10-01	Amish Chadha	Register opened with the open decisions from GGD-DDR-001, GGD-DDR-002, the review note and the design for construction (GGD-DDR-003)

GrainGuard design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

Table 1a. Open decisions.

No.	Question	Options	Recommendation
9	The 20 W panel decided on 2026-10-02 was conditional on the stay-off mast check still passing. With the stay off, the DN25 mast alone is at 0.9 times yield at a 45 m/s gust (GGD-CAL-001 F6), under the 1.5 times margin	(a) Keep the DN25 mast and allow the stay off only in gusts under 30 m/s (2.1 times yield), by a rule in the build plan (written in section 3.6 as the proposal). (b) Move to a DN32 mast (42.4 x 3.6 mm, 1.5 times margin at 45 m/s): changes the mast, base plate, six clamps and BOM line 10. (c) Go back to the 10 W panel, and R8 recovery is not met again	(a): the stay is off only for service, on a calm day

■ To confirm when parts are bought

Table 2. Items to confirm when parts are bought or the site is chosen.

#	What to confirm	Why it matters	Source
1	The bin maker rates the center hanger for cable loads, and for at least the 2.5 kN design pull-down	The rope hangs from it; an unrated roof can buckle (R10)	GGD-CAL-001 F3; build plan S3
2	The position and spacing of two stiffener bolts at a sheet seam between 1.4 and 1.8 m up, near the mast position	They set the wall bracket's holes and the stay height; the mast lug must match	GGD-DDR-003 P5
3	The peak cap is fixed (it does not lift for filling) and has room for an M12 gland 300 from the axis	The bus cable leaves through it; if the cap lifts, the cable goes through a roof panel next to it instead	GGD-DDR-003 P4
4	The enclosure's lug kit, lug spacing and inner boss positions	They set the mounting plate's lug holes and the battery shelf position	GGD-DDR-003 P6
5	The solar panel (20 W, decided 2026-10-02) has a frame with a flat back lip at least 12 wide, and its size fits the panel bracket	The panel bolts through it; the bracket was drawn for the 10 W panel and is now 430 mm wide for a panel of about 430 x 350 mm	GGD-DDR-003 P7; decision of 2026-10-02 (open item 2)
6	The mast clamps' V-saddles fit 33.7 mm (DN25) pipe and take 3 to 4 mm plate	Every part on the mast hangs on them	GGD-DDR-003 P5 to P8
7	The charge controller can clamp its temperature compensation at 15.0 V, and has an adjustable low-voltage disconnect that can be set to the battery maker's 50 % figure at -20 °C (about 12.1 V)	Keeps the system at 15 V or less in the cold (R12) and the battery from freezing (R9)	GGD-CAL-001 D8 and G2; decision of 2026-10-02 (open item 6)
8	The fan transition top is sheet steel thin enough for an M12 gland, with room 300 out from the wall clear of access panels	The plenum probe goes there	GGD-DDR-002; build plan step 14
9	The antenna's connector matches the board's, and the coax loss is 1 dB or less at 915 MHz	The radio margin assumes 1 dB of cable loss at each end (R7)	GGD-CAL-001 E1

■ Value engineering

Value-engineering target: USD 260 (a hypothetical control target, not a limit; `budget_usd` in `project.yaml`). Estimated cost of the constructable design: USD 406.50 per bin (USD 146.50 over the target), including the 20 W panel (USD 34.00, USD 9.00 more) and the adjustable-disconnect charge controller (USD 18.00, USD 8.00 more) decided on 2026-10-02, with the farmhouse receiver (USD 22.00) costed once per farm. The concept kit was USD 256.50; making it buildable added USD 133.00 and the 2026-10-02 decisions USD 17.00 (GGD-DDR-003, GGD-CAL-001 H1 and H4).

Main cost drivers:

- The mast weldment, stay, wall bracket, anchors and six mast clamps: USD 53.00 (the concept costed USD 15.00 for a mast with no joints).
- The six pods: USD 67.00 together, of which the glands, gasket, screws and rope stops added for construction are USD 21.00.
- The relay kit with its post mount: USD 39.00; the radio and bus board: USD 30.00.

- Three lines missing from the concept: antenna USD 15.00, battery fuse and surge protection USD 12.00, conduit USD 12.00.

Savings worth trying:

- Mount the controller box, panel and shield on the wall stiffener instead of a mast: saves roughly USD 40 (mast weldment, anchors, stay and two clamps), but the panel's facing is then set by which side of the bin the stiffener is on, and the ambient sensor sits close to the wall's reflected heat, so it changes the concept and would need Amish's decision.
- Fit the antenna directly on the box with a sealed bulkhead: saves about USD 8; the radio margin falls to about 12 dB at 1 km, still meeting R7.
- Crimped ferrule stops instead of set-screw rope stops: about USD 10 for six, where a swaging tool is available.
- Direct-burial-rated cable clipped along the wall foot instead of conduit: about USD 8.
- Buy cable, clamps and fixings for several bins at once: the per-bin cost of lines 3, 10 and 13 falls with quantity.

■ Decisions made

Table 3. Decisions made.

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items D1 to D11: budget per bin with the receiver per farm and cost trims, pitch reword, control rule and defaults, RH-based moisture sensing, one center cable with six pods, RS-485 pod bus, solar controller with an electrician-installed relay kit, AGM battery, plenum sensing if affordable, CO ₂ as an option, corn in a small Midwest bin	Amish: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them."	GGD-DDR-001
2026-09-25	Plenum temperature probe adopted into the base kit; relay input of 3 mA or less; charger clamped at 15.0 V	Amish: "i accept all your recommendations, go with them across all repos."	GGD-DDR-002, N1 to N3
2026-09-26	Budget set to USD 260 to cover the priced BOM	Amish: "i approve all the budget items."	GGD-DDR-002 v0.2
2026-09-27	Solar panel faces away from the bin, out of the wall's shade	Amish: "resolve the challenges for ConePro, BridgePulse, Grainguard and WellSense."	GGD-DDR-002 v0.3
2026-09-30	Make the design physically buildable while drawing the build plan; changes recorded as Draft for review (accepted on 2026-10-02, below)	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations."	GGD-DDR-003
2026-10-01	<code>budget_usd</code> is a hypothetical value-engineering target, not a limit; cost is reported against it	Amish: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens. its ok to ensure wording reflects that the hypothesis budget was x - the real cost being accrued is y"	This register; GGD-CAL-001 v0.4
2026-10-02	Open item 1: design for construction accepted: GGD-DDR-003 as drafted	Amish: "i approve your recommendations for all 555 open decisions."	GGD-DDR-003
2026-10-02	Open item 2: battery and panel capacity for the recovery half of R8: fit a 20 W panel, provided the mast check with the stay removed is rerun for the larger panel and still passes	Amish: "i approve your recommendations for all 555 open decisions."	GGD-DDR-001 O2; GGD-DDR-002
2026-10-02	Open item 3: the pitch says only "cool it or dry it" (no "without rewetting it"), and the cool-mode allowance is kept	Amish: "i approve your recommendations for all 555 open decisions."	GGD-DDR-002 O4
2026-10-02	Open item 4: first host farm and electrician: recruit the host farm and its electrician through a land-grant extension grain-storage specialist, for example at Purdue University or Iowa State University, choosing a farm with a small corn bin,	Amish: "i approve your recommendations for all 555 open decisions."	GGD-DDR-001 O1

Date	Decision	Decided by	Record
	an aeration fan and a center hanger. This is the route to the first candidate, not an agreed partner		
2026-10-02	Open item 5: phosphine protection: conformal coating for the prototype, with a coated test board left in the bin through one fumigation at TRL 4; move to sealed connectors if that board shows copper corrosion	Amish: "i approve your recommendations for all 555 open decisions."	GGD-PRC-001, open questions; R9
2026-10-02	Open item 6: a charge controller with an adjustable low-voltage disconnect, set to the battery maker's 50 % figure at -20 °C (about 12.1 V); confirmed at TRL 4	Amish: "i approve your recommendations for all 555 open decisions."	GGD-CAL-001 G2; R9
2026-10-02	Open item 7: the appearance model is brought into line with GGD-DDR-003 and re-rendered on Amish's Mac at the next render session	Amish: "i approve your recommendations for all 555 open decisions."	Review note 2026-09-26, items 1, 4, 5 and 6; GGD-DDR-003
2026-10-02	Open item 8: the control rule defaults and target moistures are confirmed with the extension specialist of item 4 and the host farmer before automatic mode is first used	Amish: "i approve your recommendations for all 555 open decisions."	GGD-PRC-001, open questions